



HWA CHONG INSTITUTION
C2 Preliminary Examination
Higher 2

NAME

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CT GROUP

16S

CHEMISTRY

9729/04

Paper 4 Practical

25 August 2017

2 hours 30 minutes

Candidates answer on the Question Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Give details of the practical shift and laboratory where appropriate, in the boxes provided. Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

Qualitative Analysis Notes are printed on pages 15 and 16.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
4	
Total	

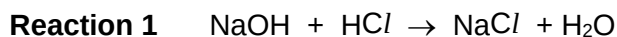
Answer **all** the questions in the spaces provided.

1 Determination of concentrations of sodium hydroxide, and of sodium carbonate in a mixture

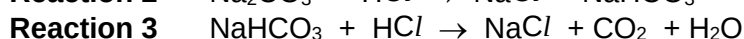
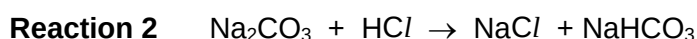
FA 1 is a solution containing sodium hydroxide, NaOH, and sodium carbonate, Na₂CO₃.

FA 2 is an aqueous solution containing 0.90 mol dm⁻³ of hydrochloric acid, HCl.

Sodium hydroxide reacts with hydrochloric acid according to the equation below:



Sodium carbonate reacts with hydrochloric acid in two separate stages. The reactions that occur are:



You are required to find the concentrations of sodium hydroxide, and of sodium carbonate, in **FA 1**, by means of a *double-indicator* titration.

In a *double-indicator* titration, **two different** indicators are used, separately, in the **same** titration. In this experiment, thymolphthalein indicator, followed by methyl orange indicator, will be used.

Thymolphthalein indicates the end-point when **Reaction 1** and **2** are complete, while methyl orange indicates the end-point when **Reaction 3** is complete.

(a) (i) Dilution of FA 2

Using a burette, measure between 35.00 cm³ and 36.00 cm³ of **FA 2** into the 250 cm³ volumetric flask.

Record your burette readings and the volume of **FA 2** added to the flask in the space below.

Make up the contents of the flask to the 250 cm³ mark with deionised water. Place the stopper in the flask and mix the contents thoroughly by slowly inverting the flask a number of times. This solution is **FA 3**.

(ii) Titration of FA 1 against FA 3

Fill a second burette with **FA 3**.

Pipette 25.0 cm³ of **FA 1** into a conical flask.

Replace the cap over the FA 1 bottle to prevent absorption of carbon dioxide from the atmosphere.

Add a few drops of thymolphthalein indicator and titrate **FA 1** with **FA 3**. The end-point is reached when the solution turns colourless. Ignore any cloudiness that you may observe in the conical flask. Record your titration results in the space below. **The volume of FA 3 used to reach the first end-point need not be consistent.**

Do not discard this solution.

To **this** solution, add a few drops of methyl orange indicator and **continue** to titrate with **FA 3** until the second end-point is reached. Record your titration results in the space below.

Perform sufficient titrations to obtain accurate results for the **second end-point**, which refers to the **total** volume of **FA 3** required for the whole titration.

Make certain that all your recorded results show the precision of your working.

[3]

(b) From your titrations, obtain suitable volumes of **FA 3** for the:

- first end-point
- second end-point.

Show clearly how you obtained these volumes.

volume of **FA 3** for first end-point =

volume of **FA 3** for second end-point =

[3]

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- (c) (i) Calculate the concentration, in mol dm^{-3} , of HCl in **FA 3**.

concentration of HCl in **FA 3** = [1]

- (ii) Calculate the amount of sodium carbonate, Na_2CO_3 , present in 25.0 cm^3 of **FA 1**.

amount of Na_2CO_3 in 25.0 cm^3 of **FA 1** = [1]

- (iii) Calculate the amount of sodium hydroxide, NaOH , present in 25.0 cm^3 of **FA 1**.

amount of NaOH in 25.0 cm^3 of **FA 1** = [2]

- (iv) Use your answers from (c)(ii) and (c)(iii), calculate the concentrations, in mol dm^{-3} , of Na_2CO_3 and NaOH in **FA 1**.

concentration of Na_2CO_3 in **FA 1** =

concentration of NaOH in **FA 1** = [1]

- (d) The maximum error in a single burette reading is $\pm 0.05 \text{ cm}^3$.

When making up the diluted acid, **FA 3**, a student recorded that 35.00 cm^3 of **FA 2** was used. What are the smallest and largest possible volumes of acid that were run into the volumetric flask?

smallest volume used = cm^3

largest volume used = cm^3

[1]

- (e) A student suggested doing the titration in (a)(ii) differently – **FA 3** is placed in the conical flask and **FA 1** in the burette, using methyl orange indicator followed by thymolphthalein indicator.

Explain if this method will allow you to determine the concentrations of NaOH and Na_2CO_3 in **FA 1**.

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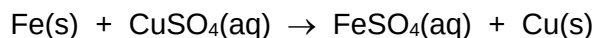
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.....[1]

[Total: 13]

2 Heat of reaction between copper(II) ions and iron metal

In this experiment, you will measure the heat given out by the reaction of excess iron with copper(II) sulfate solution and use this to find the concentration of the copper(II) sulfate.



FA 4 is iron powder.

FA 5 is aqueous copper(II) sulfate, CuSO_4 .

In an appropriate format in the space below, prepare a table in which you may record each temperature and the time it was taken.

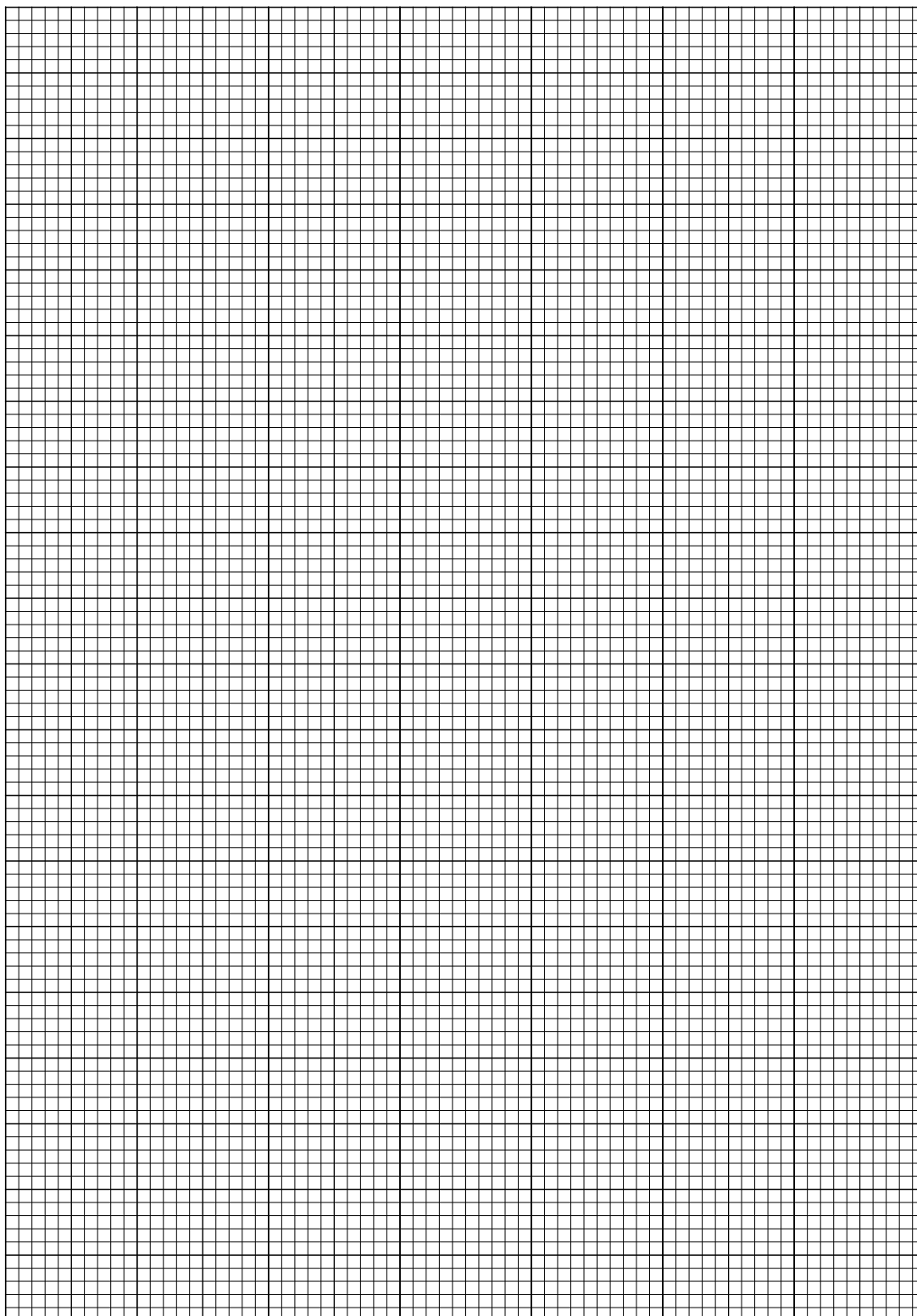
1. Wash thoroughly a burette and fill it with **FA 5**. Place a Styrofoam cup into a 250 cm³ beaker to prevent it from tipping over. Transfer 40.00 cm³ of **FA 5** into the Styrofoam cup.
2. Place the lid onto the cup and insert the thermometer through the lid. Measure and record the initial temperature of the **FA 5** solution in the cup.
3. Start the stopwatch. Measure and record the temperature of the solution in the cup **every half minute** up to and including the temperature at 1.5 min. Stir the solution using the thermometer.
4. At time $t = 2.0$ min, add all the powdered iron **FA 4** to the solution. Stir the mixture **thoroughly** with the thermometer to **ensure the solid mixes well with the solution**.
5. Record the temperature of the mixture **every minute** from $t = 2.5$ min. Continue stirring **thoroughly** and mixing the contents of the cup well **throughout** your recordings.
6. Once the temperature starts to drop, continue recording **every half minute** for a **further 3 minutes**. Constantly stir the solution **thoroughly**.

(a) Experimental Results

- (b) (i) Plot on the grid below, a graph of the temperature on the y-axis, against time, t , on the x-axis. The scale for the temperature axis must allow you to plot a point with temperature $5\text{ }^{\circ}\text{C}$ greater than the maximum temperature you recorded.

Draw the following **best-fit** straight lines on the graph.

- a line through the points before addition of **FA 4**.
- a line through the points once temperature starts to drop.



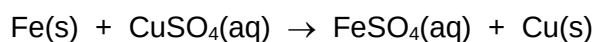
- (ii) Use the best-fit straight lines to determine the theoretical temperature change at time $t = 2.0$ min.

change in temperature = [5]

- (c) (i) Use your answer to (b)(ii) to calculate the heat energy produced in the reaction. (Assume that **4.2 J** are required to increase the temperature of 1 cm^3 of solution by 1°C .)

heat energy produced = [1]

- (ii) The molar enthalpy change, ΔH , for the reaction shown below is -152 kJ mol^{-1} .



Use this value and your answer to (c)(i) to calculate the concentration of copper(II) sulfate, in mol dm^{-3} , in **FA 5**.

concentration of copper(II) sulfate = [5]

- (d) (i) Calculate the maximum percentage error in the highest temperature that you recorded in your results table.

maximum percentage error = [1]

- (ii) What would be the expected change in temperature obtained in (b)(ii) if the volume of copper(II) sulfate used was halved? Explain your answer.

.....

[1]

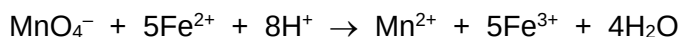
[Total: 17]

3 Planning

The concentration of copper(II) sulfate in **FA 5** in Question 2 may be determined instead, by titration of the aqueous solution obtained after step 5, with aqueous potassium manganate(VII).

The reaction mixture obtained after step 5 in Question 2 contains $\text{FeSO}_4(\text{aq})$ and solids iron and copper. This mixture needs to be first filtered. The filtrate is then diluted to obtain a solution suitable for titration with aqueous potassium manganate(VII) of known concentration, in the presence of acid.

The manganate(VII) ions react with iron(II) ions as shown in the following equation.



- (a) Using information from Question 2, determine a suitable concentration of aqueous potassium manganate(VII) to use in this titration. The concentration of manganate(VII) should not exceed $0.0500 \text{ mol dm}^{-3}$.

You are to assume that the concentration of copper(II) sulfate in **FA 5** is $0.800 \text{ mol dm}^{-3}$. (Note that this is not the correct value.)

Show clearly your calculations and any assumptions.

[3]

- (b)** Plan an experiment to determine the concentration of copper(II) sulfate in **FA 5** using the titration method described above.

In your plan, you should include details of:

- the quantities of chemicals you would use,
- the apparatus you would use and the procedure you would follow,
- the measurements you would take.

[5]

[Total: 8]

4 Qualitative analysis

In this question you will perform a series of test-tube reactions and use the observations to help you deduce the identities of two solids present in a given mixture **FA 6**.

You will also **devise a plan**, consisting of test-tube reactions, and **carry out** the plan to distinguish between three solutions **FA 9**, **FA 10** and **FA 11**, so that each is identified.

At each stage of any test you are to record details of the following:

- details of colour changes and precipitates formed
- the names of gases evolved and details of the test used to identify each one

You should indicate clearly at what stage in a test a change occurs.

No additional tests for ions present should be attempted.

- (a) **FA 6** is a mixture of two solids: **FA 7**, which is soluble in water and **FA 8**, which is insoluble in water. Each contains one cation and one anion listed in the Qualitative Analysis Notes on pages 15 and 16.

Carry out the following tests and record your observations in the table.

<i>test</i>		<i>observations</i>
(i)	Place all of the solid, FA 6 , into a boiling tube. Add 10 cm ³ of deionised water and shake to dissolve FA 7 . Filter the mixture, collecting the filtrate in a test-tube. Keep the filtrate for tests (ii) to (iv). Wash the residue, FA 8 , with deionised water. Collect the washings in the previous boiling tube. Keep the residue for tests (v) to (vii).	
(ii)	To a 1 cm depth of filtrate, FA 7 , in a test-tube, add aqueous sodium hydroxide. Carefully warm the mixture.	
(iii)	To a 1 cm depth of filtrate, FA 7 , in a test-tube, add aqueous ammonia.	
(iv)	To a 1 cm depth of filtrate, FA 7 , in a test-tube, add 5 drops of aqueous silver nitrate, followed by aqueous ammonia.	

<i>test</i>		<i>observations</i>
(v)	Transfer half a spatula of the residue, FA 8, into a clean boiling tube. Using a measuring cylinder, measure out 10 cm³ of dilute nitric acid. Transfer the acid to the boiling tube in portions. Mix the contents of the boiling tube thoroughly. Filter if necessary. This solution is FA 12.	
(vi)	To a 1 cm depth of FA 12 in a test-tube, add 1 cm depth aqueous edta.	
(vii)	To a 1 cm depth of FA 12 in a test-tube, add 1 cm depth aqueous sodium hydroxide, followed by aqueous ammonia. Do not discard the remaining FA 12. Keep the solution for (b).	

[4]

(viii) From the observations, identify the ions in **FA 7** and **FA 8**.

FA 7 contains the cation and the anion

FA 8 contains the cation and the anion

[2]

(ix) Explain in terms of the chemistry involved, your observations in **(a)(iv)**.

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..... [2]

(x) State the type of reaction that occurred in **(a)(vi)**.

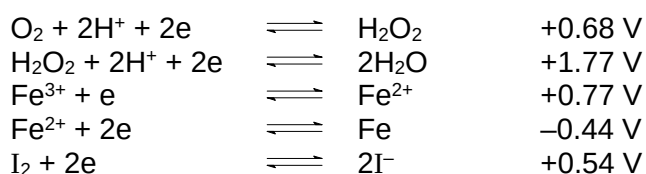
..... [1]

hydrogen peroxide iron(II) sulfate potassium iodide

You should commence your plan by adding **FA 12** to each of **FA 9**, **FA 10** and **FA 11**.

You should aim to use the **minimum number** of reactions.

You may find it useful to consider the standard electrode potentials given below.



You may use the space below to plan your tests.

In the table in the next page, record

- details of the tests, including quantity of solutions used,
- observations of the tests,
- identities of the products that gave rise to the observations,
- identity of each solution.

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings present.

	<i>observations, identities of products and identity of each solution</i>		
<i>tests</i>	FA 9	FA 10	FA 11

ConclusionSolution **FA 9** containsSolution **FA 10** containsSolution **FA 11** contains

[8]

[Total: 17]

Qualitative Analysis Notes

[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	reaction with	
	NaOH(aq)	NH ₃ (aq)
aluminium, Al ³⁺ (aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH ₄ ⁺ (aq)	ammonia produced on heating	-
barium, Ba ²⁺ (aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca ²⁺ (aq)	white ppt. with high [Ca ²⁺ (aq)]	no ppt.
chromium(III), Cr ³⁺ (aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu ²⁺ (aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe ²⁺ (aq)	green ppt. turning brown on contact with air insoluble in excess	green ppt. turning brown on contact with air insoluble in excess
iron(III), Fe ³⁺ (aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg ²⁺ (aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn ²⁺ (aq)	off-white ppt. rapidly turning brown on contact with air insoluble in excess	off-white ppt. rapidly turning brown on contact with air insoluble in excess
zinc, Zn ²⁺ (aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of aqueous anions

<i>ion</i>	<i>reaction</i>
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, $\text{Cl}^-(\text{aq})$	gives white ppt. with $\text{Ag}^+(\text{aq})$ (soluble in $\text{NH}_3(\text{aq})$)
bromide, $\text{Br}^-(\text{aq})$	gives pale cream ppt. with $\text{Ag}^+(\text{aq})$ (partially soluble in $\text{NH}_3(\text{aq})$)
iodide, $\text{I}^-(\text{aq})$	gives yellow ppt. with $\text{Ag}^+(\text{aq})$ (insoluble in $\text{NH}_3(\text{aq})$)
nitrate, $\text{NO}_3^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil
nitrite, $\text{NO}_2^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil; NO liberated by dilute acids (colourless $\text{NO} \rightarrow$ (pale) brown NO_2 in air)
sulfate, $\text{SO}_4^{2-}(\text{aq})$	gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (insoluble in excess dilute strong acids)
sulfite, $\text{SO}_3^{2-}(\text{aq})$	SO_2 liberated on warming with dilute acids; gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (soluble in dilute strong acids)

(c) Test for gases

<i>gas</i>	<i>test and test result</i>
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

<i>halogen</i>	<i>colour of element</i>	<i>colour in aqueous solution</i>	<i>colour in hexane</i>
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple

You should have the following apparatus and chemicals.

Apparatus

2 x burette (50 cm³);
 1 x pipette (25.0 cm³);
 1 x pipette filler;
 1 x retort stand and burette clamp;
 2 x filter funnel;
 1 x white tile;
 2 x 250 cm³ conical flasks;
 1 x volumetric flask (250 cm³)
 1 x Styrofoam cup;
 1 x 250 cm³ beaker;
 1 x thermometer with range –5 °C to +50 °C, graduated to 0.2 °C;
 1 x stopwatch, reading to at least 0.1s;
 1 x 10 cm³ measuring cylinder;
 1 x Bunsen burner;
 1 x lighter (shared by 2 students);
 10 x test-tube;
 2 x boiling tube;
 1 x test-tube rack;
 1 x test-tube holder;
 1 x delivery tube with rubber stopper (fits boiling tube);
 1 x marker for writing on glass;
 1 x pair of disposable vinyl gloves;
 1 x pair of safety goggles;
 1 x wash bottle containing deionised water;
 A clear plastic bag containing:
 6 x plastic dropping pipettes;
 1 x lid for Styrofoam cup;
 1 x plastic spatula;
 1 x wooden splint;
 filter papers;
 litmus papers;
 paper towels.

Chemicals

Reagents labelled FA

FA 1
FA 2
FA 4
FA 5
FA 6
FA 9
FA 10
FA 11

Bench reagents

methyl orange
 thymolphthalein
 limewater
 dilute nitric acid
 aqueous ammonia
 aqueous silver nitrate
 aqueous sodium hydroxide
 aqueous edta (ethylenediaminetetraacetic acid)